

Evaluation Protocols Change the Effective Model: A Multi-Seed Study of Rating Prediction Recommenders

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Abstract—We study how evaluation protocols alter the *effective* model in rating-prediction recommender systems. Across three datasets (ML-1M, GoodBooks, Book-Crossing) spanning a $3000 \times$ density range, we evaluate 12 models from four families (Bias, Matrix Factorization, Deep, Adaptive) under three protocols—strict cold-start (SC), warm-random (WR), and warm-temporal (WT)—over three random seeds, yielding 288 runs. We find that protocols do more than reorder rankings: they change which model components are functionally usable. Under SC, UserBias collapses exactly onto GlobalMean (difference = 0) because test users are unseen, so user-specific terms are unavailable. Protocols also favor different families: SC prefers item-bias and matrix factorization, WR prefers static collaborative models, and WT prefers behavioral, fused, and adaptive models. Rank shifts reach 10 positions and inter-protocol Kendall τ can approach zero. We conclude that a model’s effective performance is determined by protocol-induced signal availability rather than architecture alone, and propose PSI, Kendall τ , and Protocol Regret as a protocol-robustness diagnostic suite.

Index Terms—Recommender systems, evaluation protocols, cold-start, protocol sensitivity, Kendall tau, regret, multi-seed evaluation

I. INTRODUCTION

Recommendation systems are fundamental to modern information access, yet their evaluation remains surprisingly sensitive to methodological choices that are often treated as secondary. A common assumption in the literature is that more complex models will consistently outperform simpler baselines. This paper challenges that assumption through a controlled multi-dataset, multi-seed empirical study.

We investigate how three widely-used evaluation protocols affect model rankings:

- **Strict cold-start (SC)**: test users never appear in the training set;
- **Warm-random (WR)**: random row-level splitting where the same user can appear across train/test;
- **Warm-temporal (WT)**: time-ordered splitting where training precedes testing.

Across three datasets spanning a $3000 \times$ density range and 12 models from four architectural families, we observe that protocol selection does more than perturb rankings—it changes which model components are functionally usable. Most strikingly, under strict cold-start the UserBias model collapses *exactly* onto GlobalMean (RMSE difference = 0.00000 on all three datasets), because test users are unseen and user-specific biases cannot be estimated. This is a mechanism-level change in the effective model, not merely a ranking fluctuation.

Our contributions include: (1) a controlled multi-seed empirical study (288 runs) demonstrating that protocols change the effective model across three datasets and 12 models; (2) the notion of *protocol-induced functional collapse*, where protocol-driven data availability disables model components; (3) evidence of systematic model-family $\times$ protocol interaction, with SC favoring item-bias/MF, WR favoring static collaboration, and WT favoring behavioral/fused/adaptive models; and (4) a protocol-robustness evaluation framework combining the Protocol Sensitivity Index (PSI), inter-protocol Kendall τ , and Protocol Regret.

II. RELATED WORK

A. Rating Prediction and Collaborative Filtering

Rating prediction has been a core task in recommender systems research. Early approaches based on neighborhood-based collaborative filtering [1] gave way to matrix factorization methods [2], which decompose the user-item interaction matrix into latent factor representations. Netflix Prize spurred interest in hybrid approaches combining multiple signal sources [2], [3].

More recently, neural collaborative filtering [4] replaced inner-product similarity with learnable neural interactions. DeepFM [5] combines factorization machines with deep neural networks to capture both low-order and high-order feature interactions. These neural families represent the higher-complexity end of the model space studied in this paper.

B. Cold-Start and Warm-Start Protocols

Cold-start recommendation has been addressed through content-based methods [6], meta-learning approaches [7], and hybrid architectures that fuse collaborative and content signals [8]. The boundary between cold and warm users is inherently protocol-dependent. Under strict user-level splitting, all test users are unseen. Under row-level random splitting, the same user can appear in both training and test sets. Under temporal splitting, the system is evaluated on future behavior. Despite the prevalence of these protocols, their impact on which model components remain usable—and consequently on model rankings—has received limited systematic study.

C. Model Complexity and Evaluation Sensitivity

A growing body of work has questioned whether complex neural models reliably outperform simpler baselines. Dacrema

TABLE I. Dataset statistics.

Dataset	Users	Items	Ratings	Density	Timestamps
ML-1M	6,040	3,706	1,000,209	4.47%	Yes
GoodBooks	53,424	13,667	5,342,431	0.73%	Yes
Book-Crossing	278,858	271,379	1,149,780	0.0015%	No

TABLE II. Evaluation protocol definitions.

Protocol	Split method	Training-set users/items
Strict Cold (SC)	User-level	Test users never in train
Warm-Random (WR)	Row-level random	Same user can cross sets
Warm-Temporal (WT)	Time-ordered	Train < Val < Test

et al. [9] showed that many neural recommendation methods do not significantly outperform simple baselines when hyperparameters are properly tuned. Benchmark design has also been scrutinized—protocol choice can significantly alter conclusions about model superiority. Our work extends this line by systematically comparing protocol effects across multiple datasets, model families, and sparsity levels, and by introducing protocol-robustness diagnostics that go beyond single-protocol point estimates.

III. METHODOLOGY

A. Datasets

We evaluate on three public datasets spanning a $3000 \times$ density range:

ML-1M is a dense movie rating dataset with real timestamps. GoodBooks is a moderately sparse book rating dataset. Book-Crossing is extremely sparse, with a density three orders of magnitude lower than ML-1M. Since Book-Crossing lacks timestamps, only strict cold-start and warm-random protocols are evaluated on this dataset, giving eight valid dataset–protocol combinations.

B. Evaluation Protocols

We define three evaluation protocols that represent common practice in the literature:

C. Compared Models

We evaluate 12 models from four architectural families, ordered by increasing complexity (levels 1–4):

All models are implemented within a unified framework with consistent data loading, preprocessing, and evaluation pipelines.

D. Evaluation Metrics

Since the task is formulated as rating prediction, we use RMSE as the primary evaluation metric. We additionally define three protocol-robustness diagnostics that summarize behavior across protocols.

TABLE III. Compared models and complexity levels.

Model	Category	Complexity	Description
GlobalMean	Bias	1	Global rating average
UserBias	Bias	1	User-specific bias
ItemBias	Bias	1	Item-specific bias
UserItemBias	Bias	1	User + Item bias
SVD	MF	2	Matrix factorization
NeuMF	Deep	3	Neural collaborative filtering
DeepFM	Deep	3	FM + Deep network
BehaviorMLP	Deep	3	Behavior feature MLP
ProfileMLP	Deep	3	Profile feature MLP
Hybrid	Deep	3	Profile + Behavior fusion
DualHard	Adaptive	4	Hard cold/warm switch
DualSoft	Adaptive	4	Learned soft gating

E. Protocol Sensitivity Index (PSI)

To quantify how sensitive a model’s ranking is to protocol choice, we define the Protocol Sensitivity Index:

$$PSI_m = \max_p(Rank_{m,p}) - \min_p(Rank_{m,p}) \quad (1)$$

where $Rank_{m,p}$ denotes the rank of model m under protocol p . A higher PSI indicates greater protocol sensitivity. For Book-Crossing, which has only two protocols, PSI is computed over strict cold-start and warm-random.

F. Bias-Normalized RMSE (BNR)

To assess whether a model improves over simple bias baselines, we define the Bias-Normalized RMSE:

$$BNR_m = \frac{RMSE_m}{RMSE_{best_bias}} \quad (2)$$

where $RMSE_{best_bias}$ is the RMSE of the best-performing bias baseline under the same dataset and protocol. $BNR = 1.0$ denotes parity with the best bias baseline; $BNR < 1.0$ denotes an improvement.

G. Inter-Protocol Kendall Tau

To measure ranking consistency between two protocols on the same dataset, we compute Kendall’s τ over the 12 model ranks. $\tau = +1$ indicates identical rankings, $\tau = 0$ uncorrelated rankings, and $\tau = -1$ fully reversed rankings. Low or near-zero τ between two protocols signals that they effectively select different models.

H. Protocol Regret

To distinguish models that are robust across protocols from those that excel only under specific protocols, we define per-protocol regret:

$$Regret_{m,d,p} = RMSE_{m,d,p} - RMSE_{best,d,p} \quad (3)$$

where $RMSE_{best,d,p}$ is the lowest RMSE achieved by any model on dataset d under protocol p . We then summarize each model by its *mean regret* (averaged over available protocols) and its *worst regret* (the maximum per-protocol regret). Low mean and low worst regret indicate robust models; low mean with high worst regret indicates protocol-specialized models.

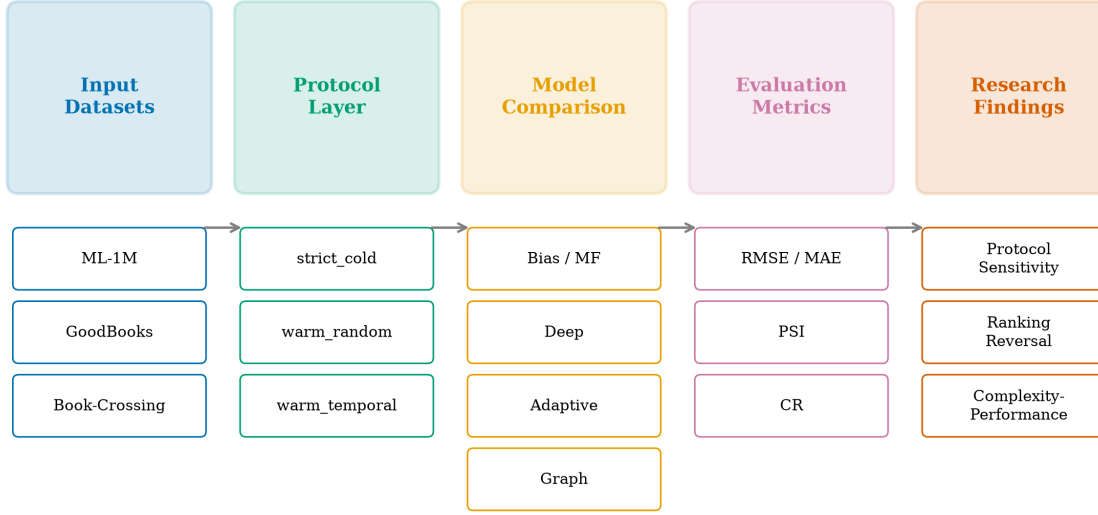
Figure 1: Research Framework

Fig. 1. Research framework. Protocols reshape the effective model by altering which signals are available to each model family under a given split; multi-seed evaluation then exposes protocol-induced functional collapse and family $\times$ protocol interaction.

I. Experimental Setup

All experiments are repeated over three random seeds (2024, 2025, 2026). We evaluate 12 models across eight valid dataset–protocol combinations: three protocols for ML-1M, three for GoodBooks, and two for Book-Crossing. The final evaluation therefore consists of $12 \times 8 \times 3 = 288$ model–protocol–seed runs. All reported RMSE values are the mean over the three seeds, with standard deviations shown in parentheses.

IV. EXPERIMENTAL RESULTS

A. Overall Performance

The overall results show that no single model family dominates across all protocols and datasets. ItemBias wins under SC on ML-1M and GoodBooks; SVD wins under WR on both datasets; DualSoft wins under WT on ML-1M while Hybrid wins under WT on GoodBooks. On the extremely sparse Book-Crossing, BehaviorMLP wins under SC and Hybrid wins under WR. The winning family therefore rotates with the protocol, motivating the finer-grained analyses below.

B. Protocol-Induced Functional Collapse

We report a mechanism-level finding: under strict cold-start, the UserBias model becomes functionally identical to GlobalMean on every dataset.

Because SC splits at the user level, no test user appears in the training set, so the user-specific bias term in UserBias has no training signal and defaults to zero. The two models therefore yield identical predictions. The effect propagates one level up: UserItemBias-SC collapses toward ItemBias-SC, with the residual user-bias contribution shrinking to differences of 0.00037–0.00493 RMSE. This is not a ranking fluctuation but a change in which model components are functionally usable—the protocol silently rewrites the effective model.

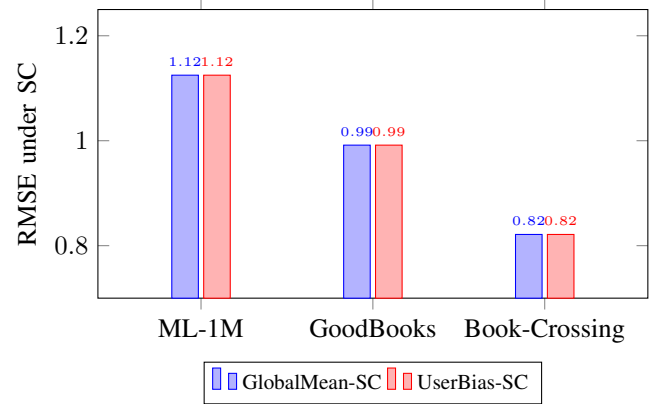


Fig. 2. Functional collapse under strict cold-start. GlobalMean-SC and UserBias-SC produce identical RMSE on all three datasets: the user-bias term is unestimable for unseen test users, so UserBias degenerates exactly to GlobalMean.

C. Protocol-Dependent Ranking Shifts

Rank shifts are systematic rather than isolated to a single model. On GoodBooks, Hybrid rises from SC rank 11 to WT rank 1, a 10-position shift; on ML-1M, BehaviorMLP shifts by 8 positions (SC rank 11 to WT rank 3); and ItemBias shifts by 7 positions on ML-1M (SC rank 1 to WR rank 8). Across all 12 models, multiple models exhibit large protocol-conditional advantages. The conclusion is that there is no cross-protocol universal best model—the winner is selected by the protocol.

D. Ranking Consistency: Kendall Tau

Inter-protocol Kendall τ values are far below 1. The most extreme case is GoodBooks SC-vs-WT with $\tau = -0.030$, i.e., the SC and WT rankings are essentially uncorrelated. Even the most consistent pair (ML-1M SC-vs-WR, $\tau = 0.606$) leaves

TABLE IV. Main results: RMSE (lower is better), mean over 3 seeds. The number in parentheses is the standard deviation in units of 10^{-3} . Bold marks the best mean in each column. SC = Strict Cold-Start, WR = Warm-Random, WT = Warm-Temporal; BC WT is excluded (no timestamps).

Category	Model	ML-1M			GoodBooks			Book-Crossing	
		SC	WR	WT	SC	WR	WT	SC	WR
Bias	GlobalMean	1.1249(8)	1.1102(2)	1.1035(0)	0.9916(2)	0.9909(1)	0.9655(0)	0.8214(7)	0.8192(1)
Bias	UserBias	1.1249(8)	1.0304(1)	1.0941(0)	0.9916(2)	0.8943(1)	0.9500(0)	0.8214(7)	0.7350(1)
Bias	ItemBias	0.9860(2)	0.9752(2)	0.9632(0)	0.9547(2)	0.9535(1)	0.9361(0)	0.8092(6)	0.8071(1)
Bias	UserItemBias	0.9896(4)	0.9056(1)	0.9544(0)	0.9596(2)	0.8535(1)	0.9198(0)	0.8088(7)	0.7237(1)
MF	SVD	0.9880(2)	0.8983(1)	0.9523(0)	0.9624(2)	0.8494(0)	0.9293(0)	0.8097(7)	0.7268(1)
Deep	NeuMF	1.0446(18)	0.9158(16)	1.0261(29)	0.9712(5)	0.8549(10)	0.9329(7)	0.8196(6)	0.7486(8)
Deep	DeepFM	1.3153(63)	1.2620(16)	1.2689(126)	1.0409(54)	1.0467(173)	0.9447(16)	0.8389(16)	0.7854(27)
Deep	BehaviorMLP	1.2570(175)	0.9180(0)	0.9262(1)	0.9791(8)	0.8594(1)	0.8616(2)	0.6774(122)	0.6570(30)
Deep	ProfileMLP	1.0889(5)	1.0596(2)	1.0652(1)	0.9803(2)	0.9793(1)	0.9562(0)	0.8158(5)	0.8109(1)
Deep	Hybrid	1.0073(4)	0.9058(0)	0.9248(1)	0.9945(8)	0.8578(1)	0.8576(1)	0.6836(126)	0.6495(28)
Adaptive	DualHard	1.0886(7)	0.9164(1)	0.9275(0)	0.9787(2)	0.8593(1)	0.8593(1)	0.7618(48)	0.7443(15)
Adaptive	DualSoft	1.0683(18)	0.9151(0)	0.9242(1)	0.9945(10)	0.8592(1)	0.8606(0)	0.6906(123)	0.6748(35)

TABLE V. Functional collapse under strict cold-start (3 seeds). GM—UB: GlobalMean minus UserBias (exactly 0 on all datasets). UIB—IB: UserItemBias minus ItemBias (near 0; on BC UserItemBias is marginally below ItemBias).

Dataset	GlobalMean	UserBias	GM—UB	UIB—IB
ML-1M	1.1249	1.1249	0.00000	0.00359
GoodBooks	0.9916	0.9916	0.00000	0.00493
Book-Crossing	0.8214	0.8214	0.00000	−0.00037

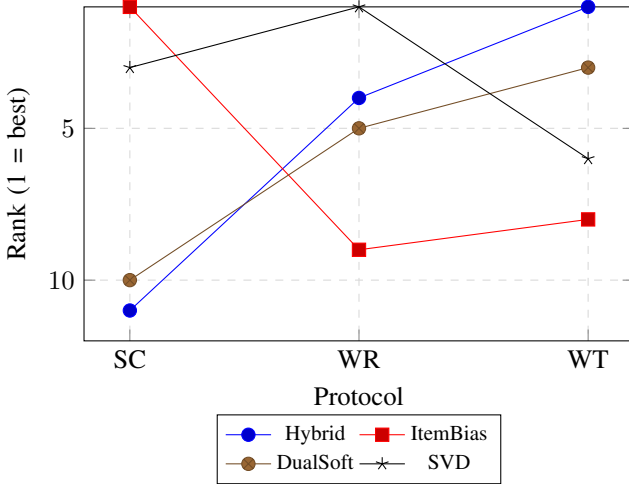


Fig. 3. Model rank across protocols on GoodBooks (3 seeds, lower is better). Hybrid rises from SC rank 11 to WT rank 1 (a 10-position shift), while ItemBias falls from SC rank 1 to WT rank 8: no model is universally best, and protocol choice determines the winner.

substantial rank disagreement. This quantifies the qualitative observation that different protocols effectively select different models.

E. Protocol Regret: Specialized vs. Robust Models

Regret separates robust from specialized models. On ML-1M, Hybrid, SVD, and UserItemBias form a robust cluster with mean regret ≤ 0.014 and worst regret ≤ 0.030 . BehaviorMLP

TABLE VI. Inter-protocol Kendall τ on model rankings (12 models, 3 seeds). Lower τ means weaker consistency; GoodBooks SC-vs-WT is essentially uncorrelated.

Dataset	Protocol pair	Kendall τ
ML-1M	SC vs. WR	0.606
ML-1M	SC vs. WT	0.303
ML-1M	WR vs. WT	0.576
GoodBooks	SC vs. WR	0.333
GoodBooks	SC vs. WT	−0.030
GoodBooks	WR vs. WT	0.455
Book-Crossing	SC vs. WR	0.576

TABLE VII. Protocol regret on ML-1M (3 seeds), sorted by mean regret. Mean regret averages over SC/WR/WT; worst regret is the maximum per-protocol regret.

Model	Mean regret	Worst regret
Hybrid	0.0098	0.0213
SVD	0.0101	0.0281
UserItemBias	0.0137	0.0302
DualSoft	0.0330	0.0823
ItemBias	0.0386	0.0769
DualHard	0.0413	0.1026
NeuMF	0.0593	0.1019
BehaviorMLP	0.0976	0.2710
ProfileMLP	0.1350	0.1612
UserBias	0.1470	0.1699
GlobalMean	0.1767	0.2119
DeepFM	0.3459	0.3636

is the clearest specialized case: its mean regret (0.0976) is moderate but its worst regret (0.2710) is large, because it ranks 11th under SC (regret 0.2710) yet 3rd under WT (regret 0.0020). DeepFM is simply weak under all protocols (mean 0.3459, worst 0.3636). The mean/worst regret pair therefore exposes a trade-off that a single-protocol ranking hides.

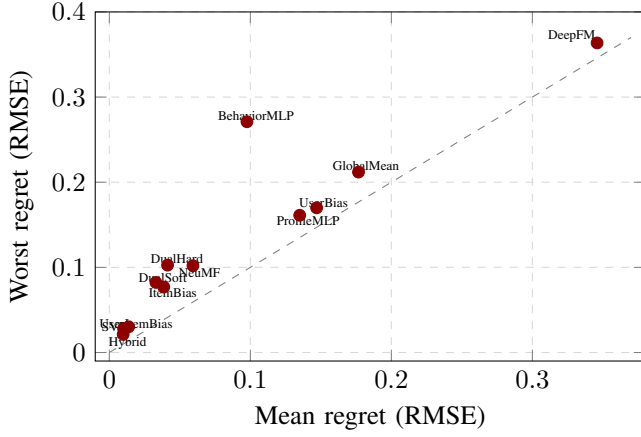


Fig. 4. Mean vs. worst regret on ML-1M (3 seeds). The bottom-left cluster (Hybrid, SVD, UserItemBias, DualSoft) is robust; points far above the diagonal, especially BehaviorMLP (mean 0.098, worst 0.271), are protocol-specialized—good on average but with a large worst-case gap.

TABLE VIII. Protocol Sensitivity Index (PSI) by dataset (3 seeds). Higher PSI = stronger protocol dependence.

Model	ML-1M	GoodBooks	BC
GlobalMean	2	4	2
UserBias	1	2	5
ItemBias	7	8	4
UserItemBias	4	3	1
SVD	4	5	2
NeuMF	3	4	1
DeepFM	0	3	3
BehaviorMLP	8	3	1
ProfileMLP	2	4	3
Hybrid	2	10	1
DualHard	3	4	3
DualSoft	5	7	0

F. Protocol Sensitivity Index

Hybrid reaches PSI=10 on GoodBooks because it ranks 11th under SC but 1st under WT, the largest single-model shift in the study. BehaviorMLP reaches PSI=8 on ML-1M (SC rank 11 to WT rank 3). At the other extreme, DeepFM has PSI=0 on ML-1M because it ranks last under all three protocols—insensitivity here reflects consistent poor performance rather than robustness. PSI therefore needs to be read alongside absolute regret.

G. Activity Gap and Model Complexity

Within the warm-random protocol, each test sample is labeled as cold, medium, or warm based on the user’s training-set activity. We define the Activity Gap as:

$$\text{ActivityGap}_m = \text{RMSE}_{\text{cold_segment}} - \text{RMSE}_{\text{warm_segment}} \quad (4)$$

A weak positive trend between complexity and the activity gap is visible: Bias models (complexity 1) cluster at gaps of 0.034–0.083, while the largest gap belongs to DualHard (complexity 4, gap 0.1884). However, the relationship is far from monotone—ProfileMLP (complexity 3) has the second-smallest gap (0.0376), and complexity-3 models span 0.038–

TABLE IX. Activity-segment RMSE gap on ML-1M warm-random (3 seeds). Gap = cold-segment RMSE – warm-segment RMSE.

Model	Cmplx.	Cold RMSE	Warm RMSE	Gap
DualHard	4	1.0981	0.9096	0.1884
BehaviorMLP	3	1.0434	0.9119	0.1315
Hybrid	3	1.0249	0.8994	0.1255
NeuMF	3	1.0301	0.9066	0.1236
DualSoft	4	1.0127	0.9093	0.1035
DeepFM	3	1.3436	1.2439	0.0997
SVD	2	0.9875	0.8918	0.0956
UserItemBias	1	0.9840	0.9009	0.0830
ItemBias	1	1.0389	0.9738	0.0651
UserBias	1	1.0676	1.0281	0.0395
ProfileMLP	3	1.0974	1.0598	0.0376
GlobalMean	1	1.1446	1.1102	0.0344

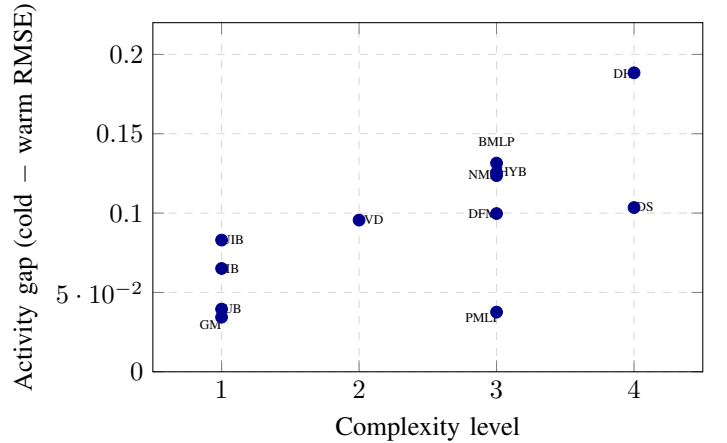


Fig. 5. Complexity vs. activity gap on ML-1M warm-random (3 seeds). Complexity shows only a weak positive correlation with the cold–warm gap; DualHard (complexity 4) has the largest gap, while several complexity-3 models span the full range, refuting a simple mapping from complexity to robustness.

TABLE X. Warm-random RMSE across datasets (3 seeds). Behavioral and hybrid models win under extreme sparsity (BC); SVD wins on denser datasets.

Model	BC (0.0015%)	GB (0.73%)	ML-1M (4.47%)
GlobalMean	0.819	0.991	1.110
UserBias	0.735	0.894	1.030
ItemBias	0.807	0.953	0.975
UserItemBias	0.724	0.854	0.906
SVD	0.727	0.849	0.898
NeuMF	0.749	0.855	0.916
DeepFM	0.785	1.047	1.262
BehaviorMLP	0.657	0.859	0.918
ProfileMLP	0.811	0.979	1.060
Hybrid	0.650	0.858	0.906
DualHard	0.744	0.859	0.916
DualSoft	0.675	0.859	0.915

0.132. Model complexity therefore does not reliably predict robustness across user activity levels.

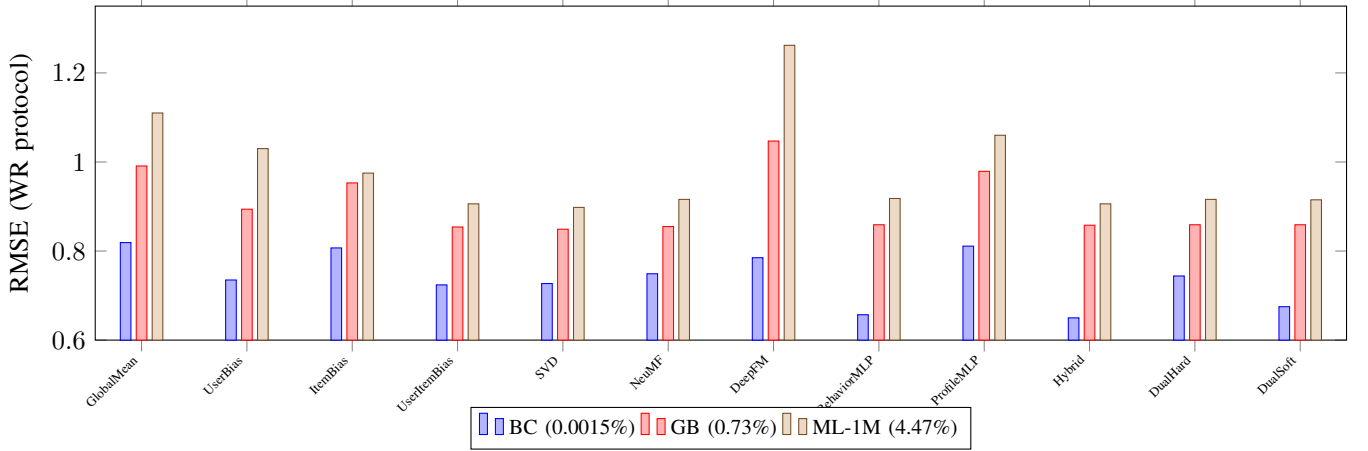


Fig. 6. Warm-random RMSE of all 12 models across the three datasets (3 seeds). Under extreme sparsity (BC), behavioral and hybrid models (BehaviorMLP, Hybrid, DualSoft) outperform all bias and MF baselines; on denser datasets (GB, ML-1M), SVD is best and the advantage of behavioral models disappears.

TABLE XI. Bias-Normalized RMSE (BNR) under warm-random (3 seeds). $\text{BNR} = \text{RMSE}_m / \text{RMSE}_{\text{best_bias}}$; values < 1.0 (bold) indicate improvement over the best bias baseline.

Model	ML-1M	GoodBooks	BC
GlobalMean	1.23	1.16	1.13
UserBias	1.14	1.05	1.02
ItemBias	1.08	1.12	1.12
UserItemBias	1.00	1.00	1.00
SVD	0.99	1.00	1.00
NeuMF	1.01	1.00	1.03
DeepFM	1.39	1.23	1.09
BehaviorMLP	1.01	1.01	0.91
ProfileMLP	1.17	1.15	1.12
Hybrid	1.00	1.00	0.90
DualHard	1.01	1.01	1.03
DualSoft	1.01	1.01	0.93

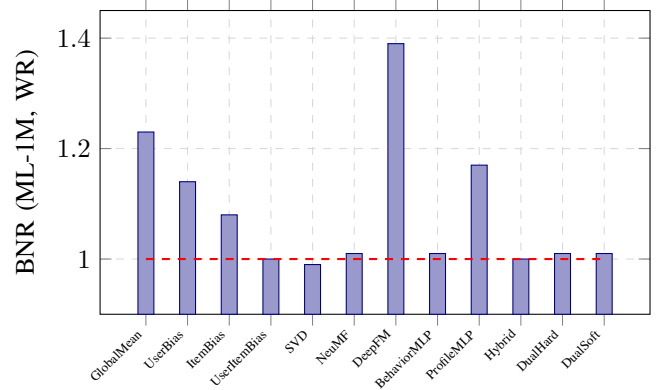


Fig. 7. BNR on ML-1M under warm-random (3 seeds). The dashed line at $\text{BNR}=1.0$ is parity with the best bias baseline. Most neural and adaptive models sit within $\pm 1\%$ of parity; only SVD marginally beats the baseline, while DeepFM is 39% worse.

H. Sparsity Effects

The warm-random results reveal that models respond very differently to sparsity. Under extreme sparsity (Book-Crossing, 0.0015% density), the behavioral and hybrid models—BehaviorMLP (0.657), Hybrid (0.650), and DualSoft (0.675)—outperform all bias and matrix-factorization baselines, indicating that profile/behavior features provide critical complementary signal when interactions are sparse. On the denser ML-1M and GoodBooks, SVD is best and these behavioral models lose their advantage. Sparsity thus interacts with model family: the relative effectiveness of behavioral fusion grows as density falls.

I. Bias-Normalized RMSE

On ML-1M and GoodBooks, most deep and adaptive models achieve BNR of 0.99–1.01, i.e., within $\pm 1\%$ of the best bias baseline, and only SVD marginally beats it ($\text{BNR}=0.99$ on ML-1M). The picture changes sharply under extreme sparsity: on Book-Crossing, BehaviorMLP (0.91), Hybrid (0.90), and DualSoft (0.93) reach BNR well below 1.0, a 7–10% RMSE reduction over the best bias baseline. BNR thus confirms that

TABLE XII. E2E vs. two-stage gating (DualSoftGating RMSE, representative run seed 2024). E2E helps only on ML-1M strict cold-start; gains elsewhere are negligible or negative.

Dataset	Protocol	2-Stage	E2E	Δ
ML-1M	SC	1.0883	1.0220	−0.0663
ML-1M	WR	0.9148	0.9117	−0.0030
ML-1M	WT	0.9238	0.9237	−0.0001
GB	SC	1.0063	1.1550	+0.1487
GB	WR	0.8592	0.8592	0.0000
GB	WT	0.8607	0.8603	−0.0004

the value of neural/behavioral modeling over bias baselines is protocol- and sparsity-dependent, not universal.

J. E2E Gating Analysis

As a supplementary analysis, we compare end-to-end trained gating with the two-stage training strategy for DualSoftGating. The values below are from a representative single run (seed 2024); the multi-seed main results for DualSoft are reported in Table IV.

E2E gating provides limited and inconsistent improvements. Although it improves performance on ML-1M strict cold-start (0.066 RMSE reduction), the gains on warm protocols are negligible (< 0.003). On GoodBooks strict cold-start, E2E actually degrades performance by 0.149. We therefore report E2E gating as an auxiliary analysis rather than a central contribution.

V. DISCUSSION

A. Protocol-Induced Functional Collapse: A Mechanism

The most consequential finding is that protocols do not merely shuffle rankings—they disable model components. Under SC, test users are unseen, so any term that depends on a per-user parameter is unestimable. UserBias therefore degenerates exactly to GlobalMean (diff=0.00000 on all datasets), and UserItemBias collapses toward ItemBias. The protocol has rewritten the effective model: a practitioner who deploys UserBias under SC is, in effect, deploying GlobalMean. This mechanism generalizes: any model whose user-specific parameters cannot be transferred to cold test users will lose its user-side capacity under SC, and the loss is silent—it appears only when one inspects which terms are functionally active.

B. Protocol Preference Structure

The three protocols favor structurally different signals:

- **SC** favors item-side priors and matrix factorization. With user terms unavailable, ItemBias and SVD win on ML-1M and GoodBooks (ItemBias rank 1, SVD rank 2–3), because item statistics and latent item factors remain estimable.
- **WR** favors static collaborative models. The same user appears in train and test, so user-specific factors are estimable and SVD/UserItemBias dominate (SVD rank 1 on ML-1M and GoodBooks).
- **WT** favors behavioral, fused, and adaptive models. Temporal drift makes static user factors stale, and models that fuse behavior/profile signals or adaptively route between cold and warm regimes win (DualSoft rank 1 on ML-1M; Hybrid rank 1 on GoodBooks).

This family $\times$ protocol interaction is systematic, not noise: it reproduces across three seeds and three datasets, and is reflected both in the per-protocol winners and in the near-zero Kendall τ (GoodBooks SC-vs-WT = -0.030).

C. Specialized vs. Robust Models: The Regret Trade-off

Protocol Regret separates models that are robust from those that are specialized. The robust cluster (Hybrid, SVD, UserItemBias on ML-1M) has low mean *and* low worst regret; these models are safe default choices. Specialized models such as BehaviorMLP have moderate mean regret but large worst regret (0.271 on ML-1M); they are excellent on their favored protocol (WT regret 0.002) but poor elsewhere (SC regret 0.271). A practitioner optimizing for a single known protocol should pick the specialized winner; one facing an unknown or mixed deployment should prefer the robust cluster. Crucially, this trade-off is invisible to a single-protocol ranking and only surfaces when mean and worst regret are reported together.

D. BNR: When Neural Modeling Pays Off

BNR shows that on ML-1M and GoodBooks, most neural and adaptive models are within $\pm 1\%$ of the best bias baseline, and only SVD marginally beats it. The computational overhead of deep models is not justified for rating prediction on these datasets under WR. The exception is extreme sparsity: on Book-Crossing, behavioral and adaptive models (BehaviorMLP, Hybrid, DualSoft) reach BNR 0.90–0.93, a 7–10% RMSE reduction over the best bias baseline. BNR thus localizes the value of neural/behavioral modeling to sparse regimes rather than treating complexity as universally beneficial.

VI. CONCLUSION

This paper presented a controlled multi-seed empirical study of how evaluation protocols change the effective model in rating-prediction recommender systems. Across 12 models, three datasets, eight dataset–protocol combinations, and three seeds (288 runs), we found:

- 1) **Multi-protocol, multi-seed unified evaluation.** Protocols systematically reorder model rankings, with rank shifts up to 10 positions and inter-protocol Kendall τ as low as -0.030 (GoodBooks SC-vs-WT); no model is universally best.
- 2) **Protocol-induced functional collapse.** Under strict cold-start, UserBias degenerates exactly to GlobalMean (diff=0.00000) and UserItemBias collapses toward ItemBias, because user-specific terms are unestimable for unseen test users. The protocol silently rewrites the effective model.
- 3) **Model-family $\times$ protocol systematic interaction.** SC favors item-bias and matrix factorization, WR favors static collaborative models, and WT favors behavioral, fused, and adaptive models—a pattern that reproduces across seeds and datasets.
- 4) **A protocol-robustness evaluation framework.** PSI, inter-protocol Kendall τ , and Protocol Regret jointly expose protocol sensitivity, ranking inconsistency, and the specialized-vs-robust trade-off that single-protocol rankings hide.

These findings suggest that recommender-system evaluation should report results across multiple protocols and seeds, and adopt protocol-robustness diagnostics rather than relying on a single-protocol point estimate.

VII. LIMITATIONS

We acknowledge several limitations:

- 1) **Three seeds, no significance testing.** Three seeds (2024/2025/2026) are a minimum for variance estimation, not a basis for formal hypothesis testing. We therefore report effect sizes (mean $\pm$ std, rank shifts, Kendall τ , regret) rather than p -values; significance testing over more seeds is left to future work.
- 2) **Book-Crossing has no timestamps.** The Book-Crossing dataset lacks timestamp information, so only SC and WR are evaluated on it; the WT protocol is unavailable for this dataset.

- 3) **Model scope.** This study covers Bias, MF, Deep, and Adaptive families. It does not include graph-based recommender models, so the conclusions apply to the four families studied and should not be extrapolated to graph recommendation without further evidence.
- 4) **No cost accounting.** We do not record parameter counts or training time. BNR reflects only relative RMSE against a bias baseline, not computational cost-effectiveness; a full cost–benefit analysis is out of scope.

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